

HOLY HYBRID

The Last Jedi of Exchange



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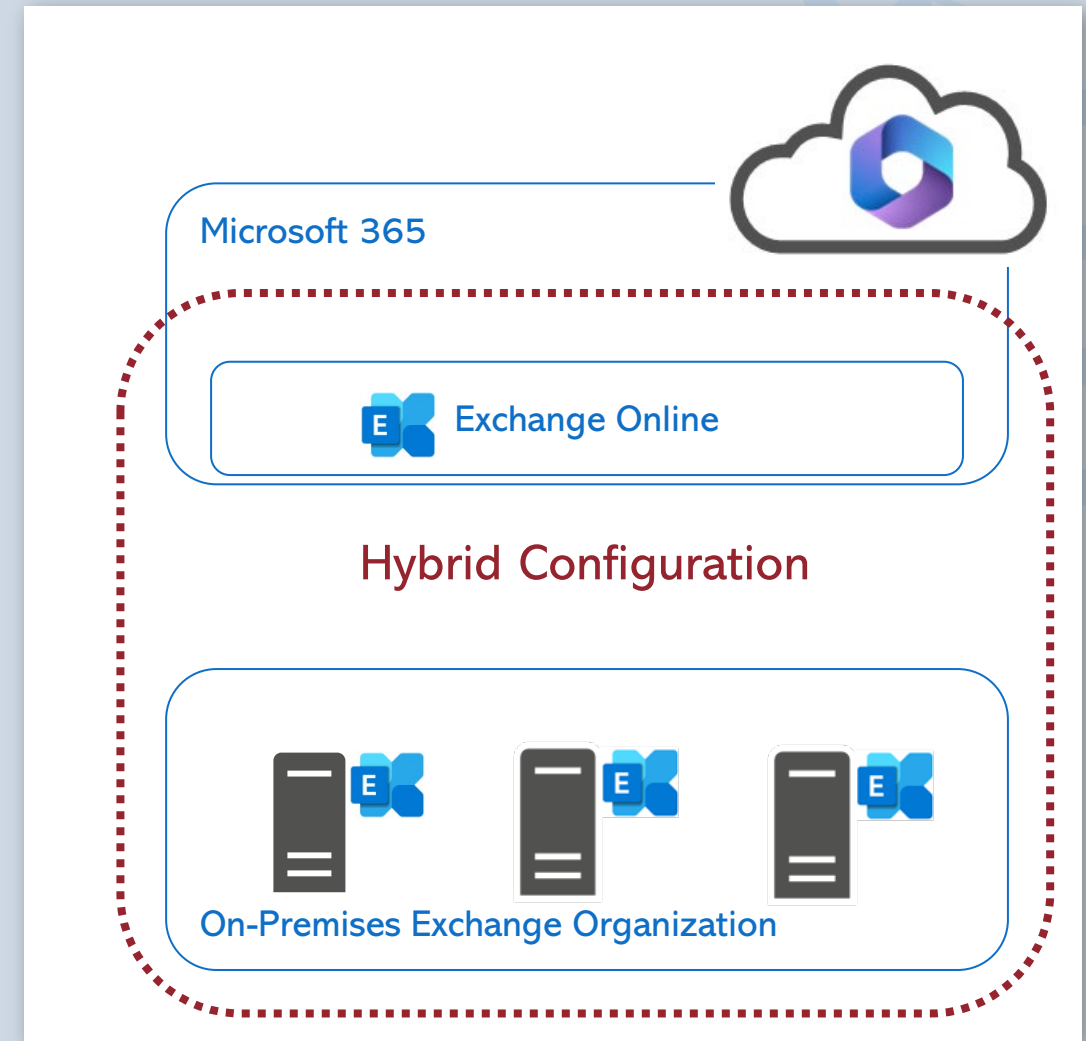
“Implementing Exchange Hybrid is simple.”

Undisclosed Exchange Consultant



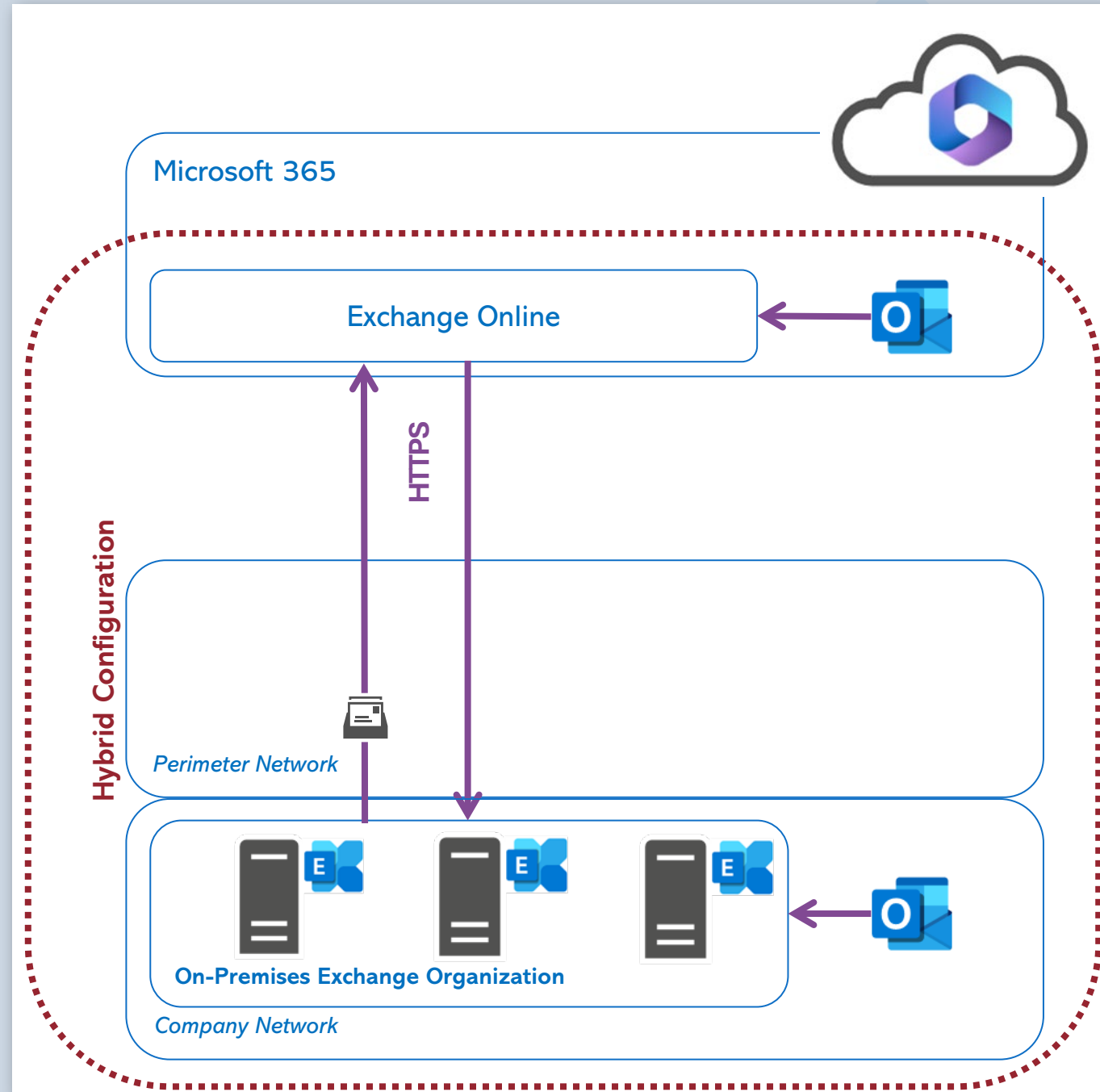
What is Exchange Hybrid?

- **Trusted** relationship between an on-premises Exchange Organization and Exchange Online
- Hybrid connections for **mail flow** (SMTP), and **service access** (HTTPS) for Exchange **hybrid functionality**
- **Hybrid Configuration Wizard (HCW)** activates and configures the hybrid mode of operation



Why do we need Hybrid?

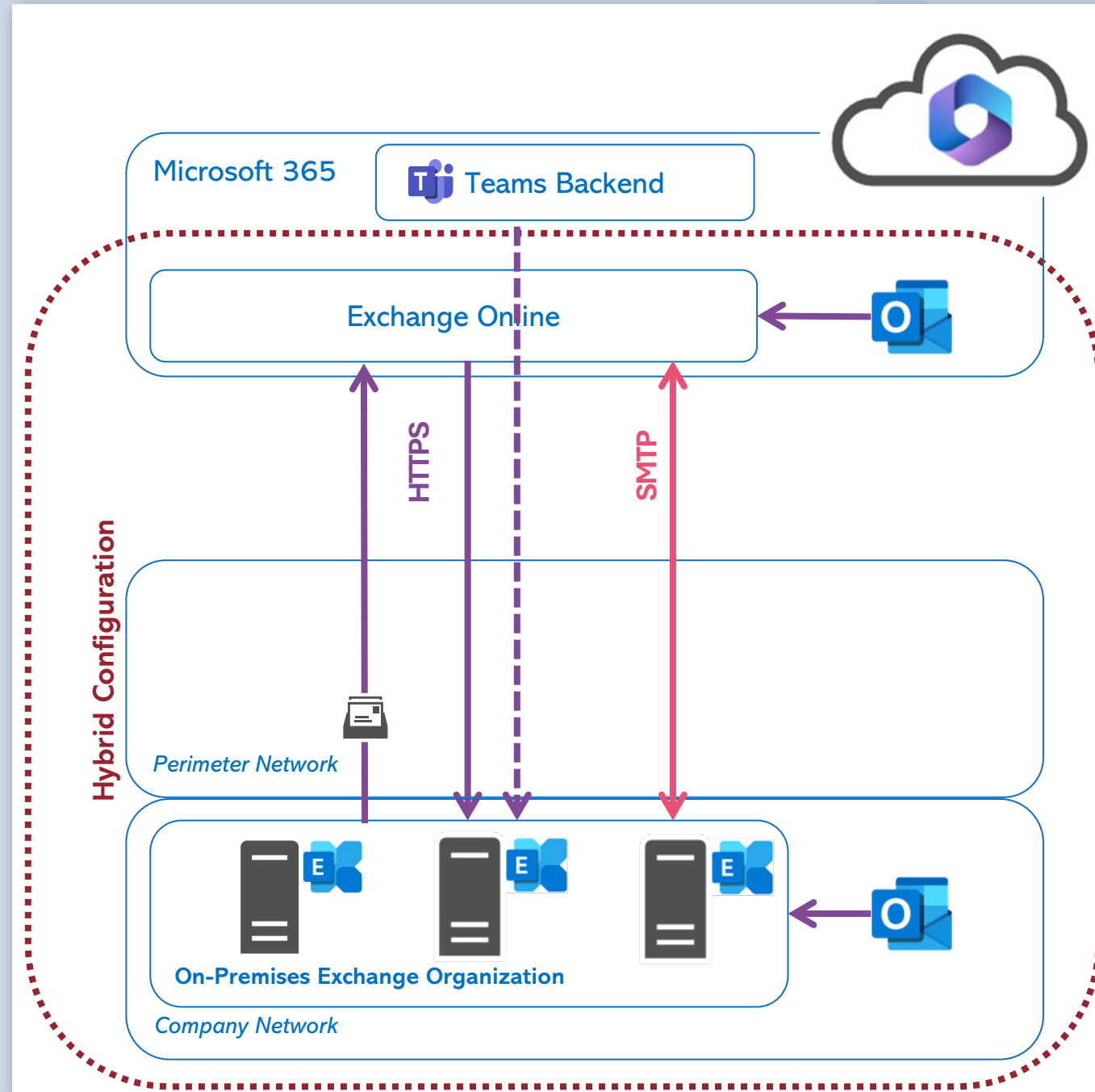
- Exchange Coexistence
 - Free/Busy Lookups, Mail Tips, Profile Pictures, etc.
- Migration
 - Native migration of mailboxes to and from Exchange Online
 - Seamless public folder migration to Exchange Online
- Transition from an On-Premises Exchange Organization to Exchange Online
 - Optimal migration experience for end users



Why do we need Hybrid?

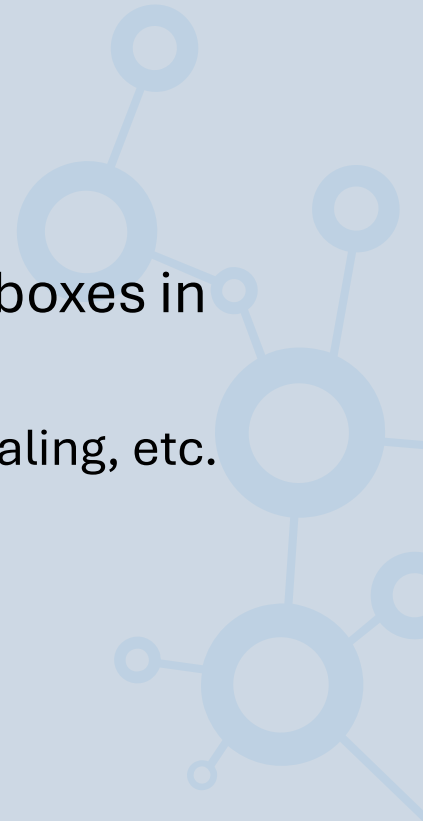
Something called Teams

- Centralized mail flow for use of on-premises email solutions with user mailboxes in Exchange Online
- Microsoft Teams with on-premises mailboxes (calendar access)



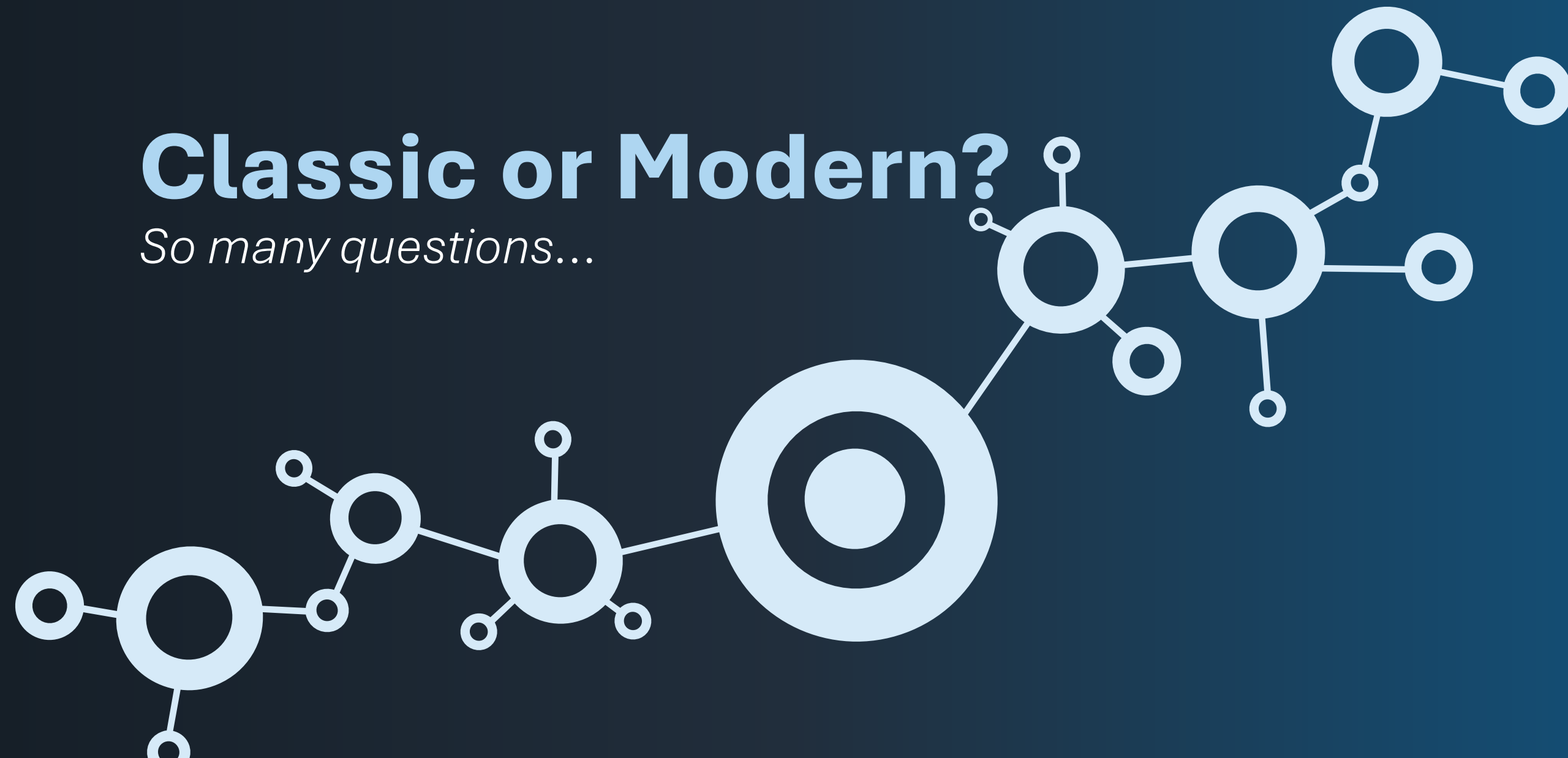
Other Reasons For Exchange Hybrid

- Centralized mail flow for use of on-premises email solutions with user mailboxes in Exchange Online
 - Gateway-based S/MIME decryption/encryption, email disclaimer, archiving, journaling, etc.
- Hybrid mail flow providing email relay functionality for on-premises legacy applications and hardware devices with
 - No direct access to the internet
 - No support for TLS protocols
 - No support for SMTPAUTH user authentication
 - No support for POP3 and IMAP4 modern authentication



Classic or Modern?

So many questions...



Exchange Hybrid

Modes of operation

Hybrid Modes

Classic

Modern

Express

Minimal

Full

Minimal

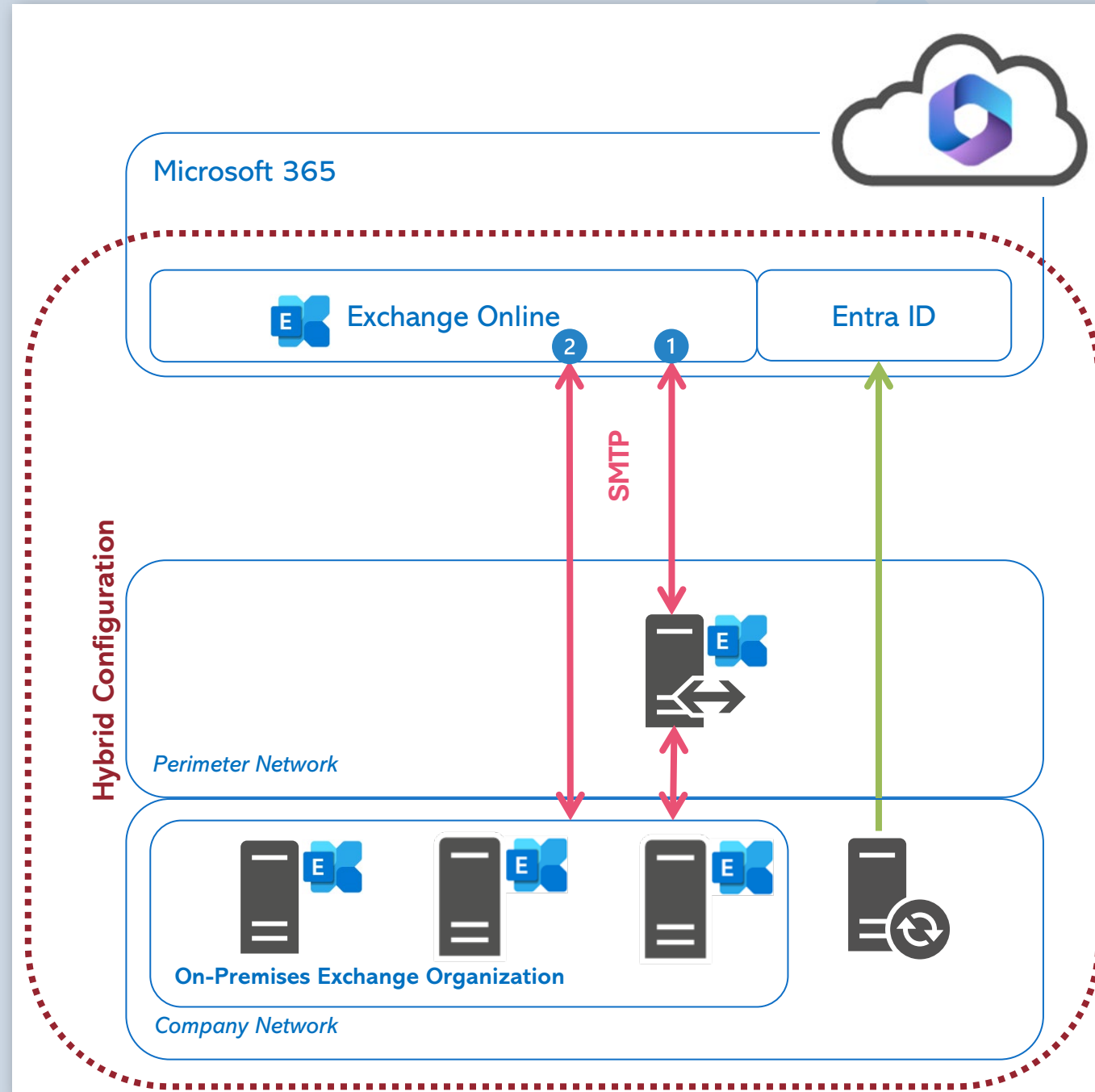
Full



Classic Full Hybrid

Mail Flow

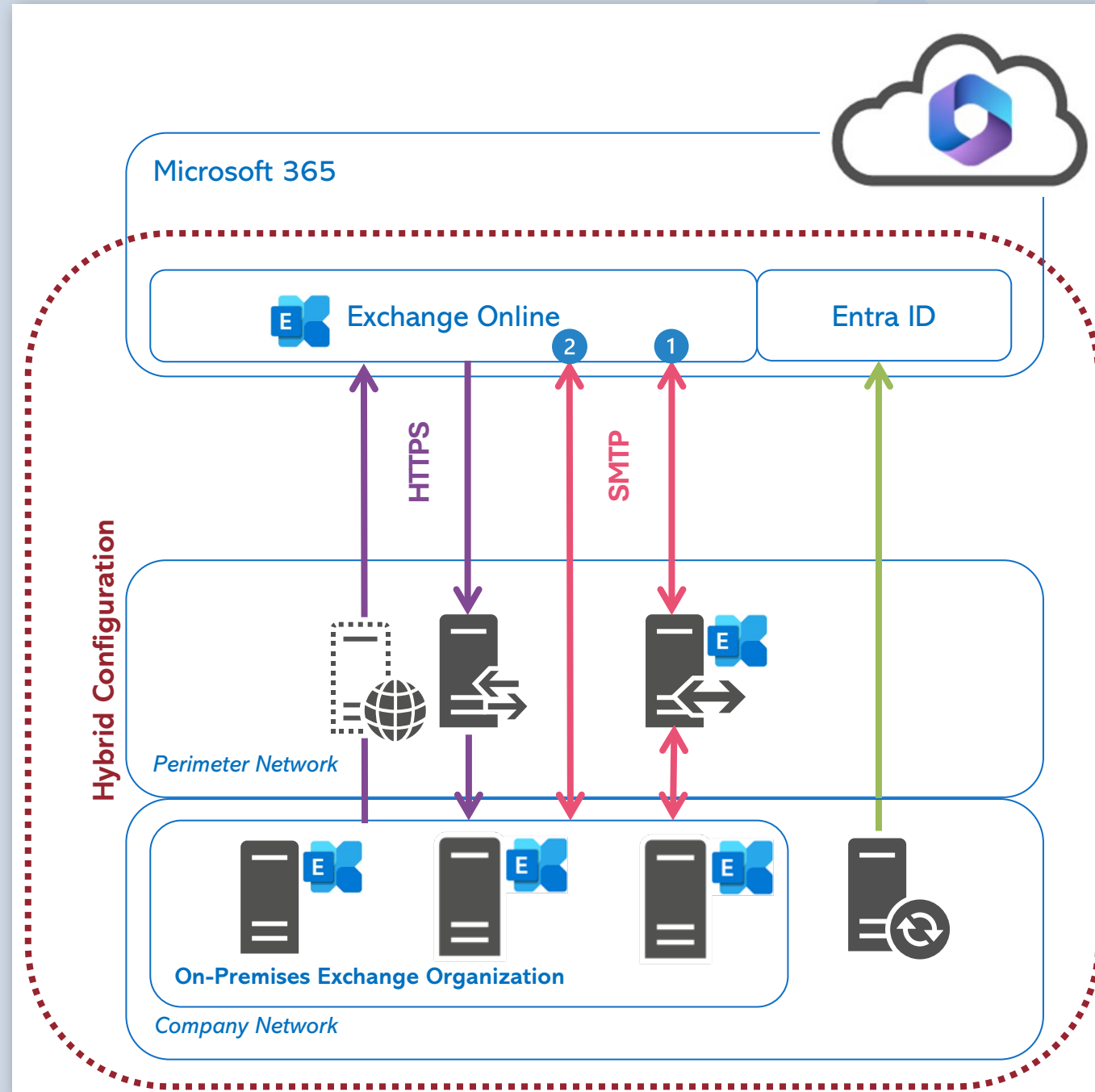
- Entra ID Hybrid with Entra ID Connect
 - Exchange Hybrid option enabled
- SMTP Connection between On-Premises and Exchange Online
 - Separate hostname for inbound SMTP connectivity (e.g., smtp365.company.com)
 - Additional public IP address
 - TLS certificate for hostname
 - Edge Transport Role in perimeter network (*Option 1*)
 - Alternatively, direct inbound connection (*Option 2*)



Classic Full Hybrid

Hybrid Functionality

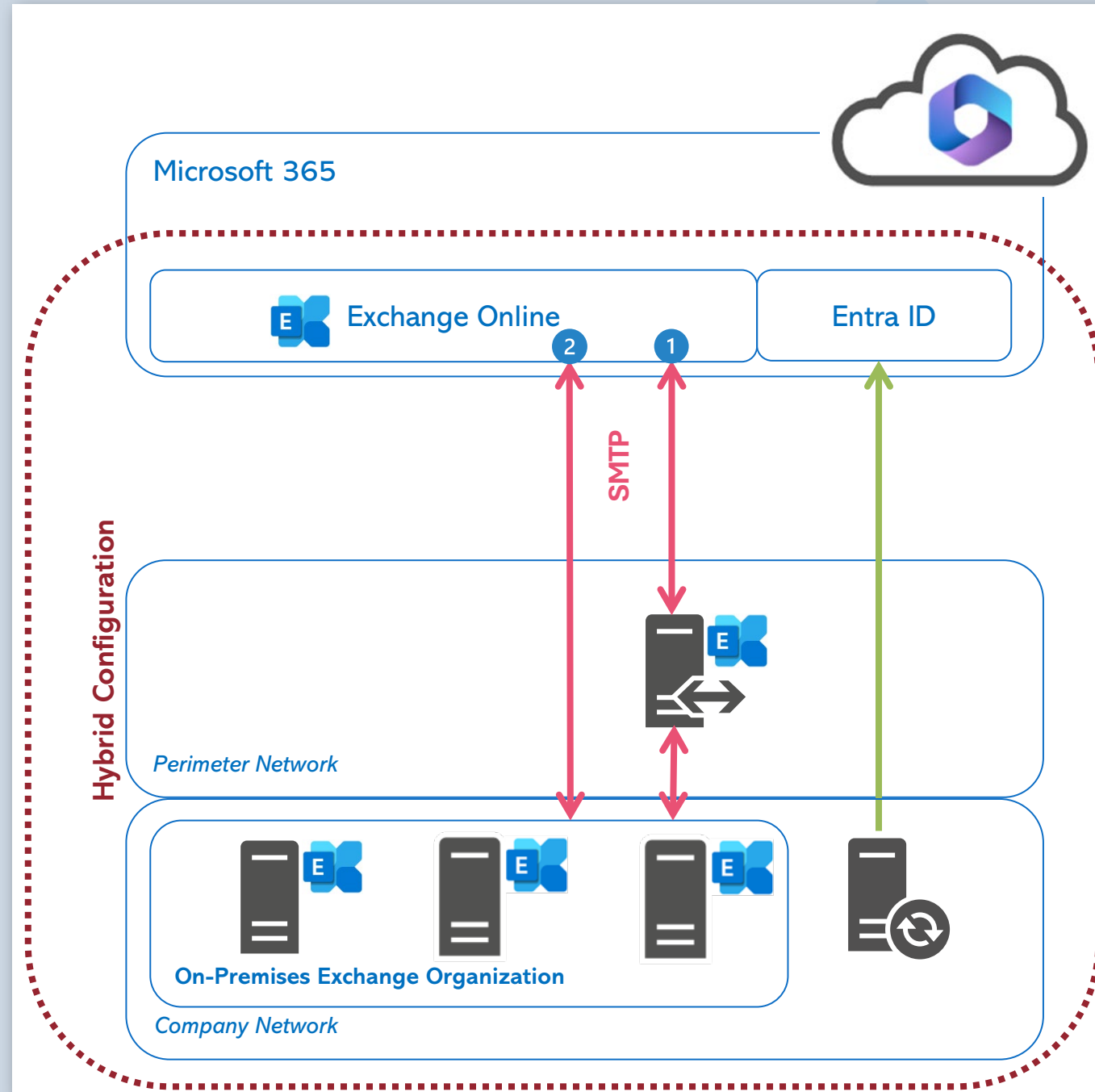
- Inbound HTTPS connection to Client Access Service
 - Internal Exchange Servers published using reverse proxy servers
 - Requires additional public IP address
- Outbound HTTPS connections to Exchange Online
 - Exchange Server communication to Exchange Online
 - Either direct or using outbound roxy server



Modern Full Hybrid

Mail Flow

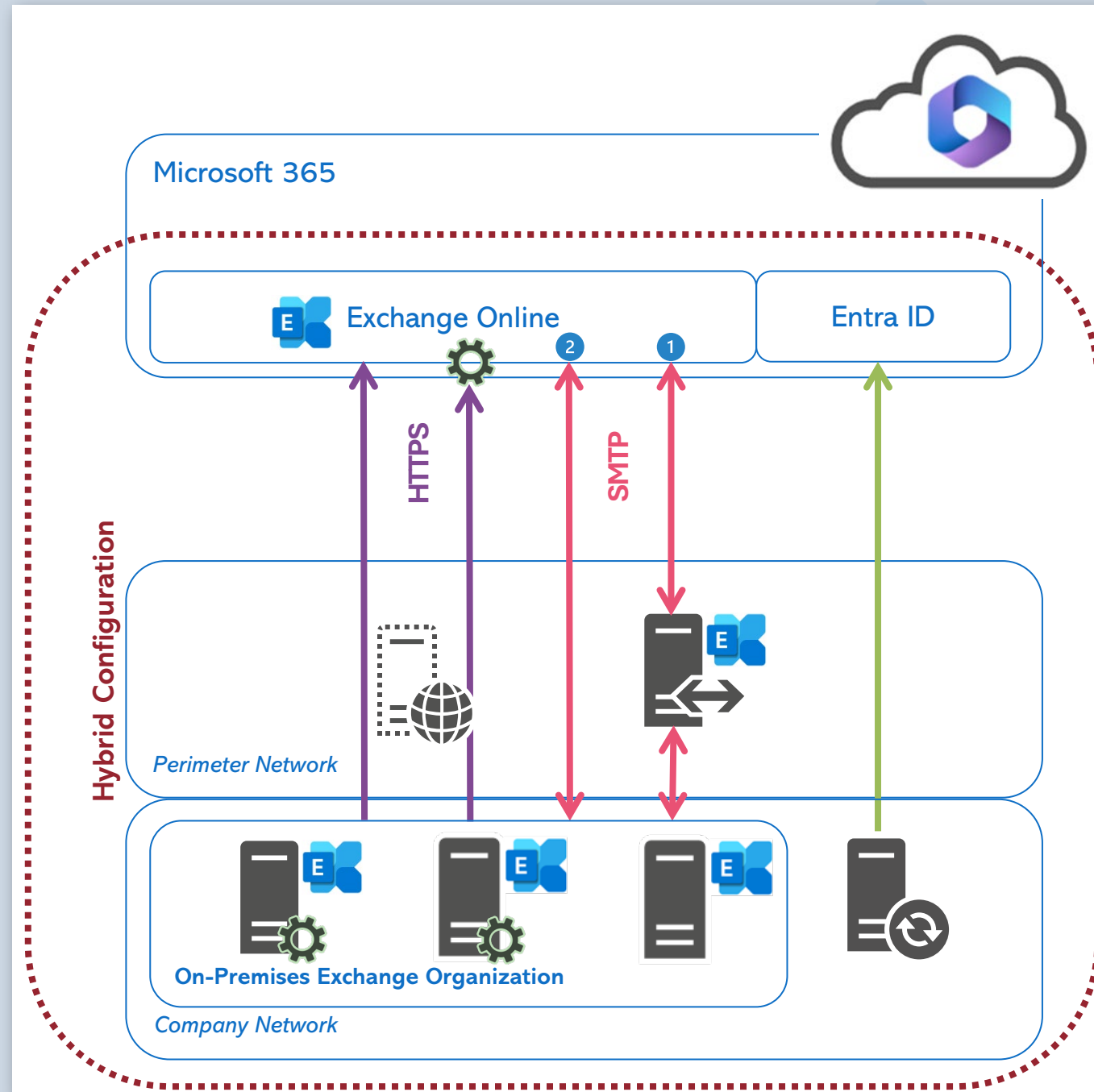
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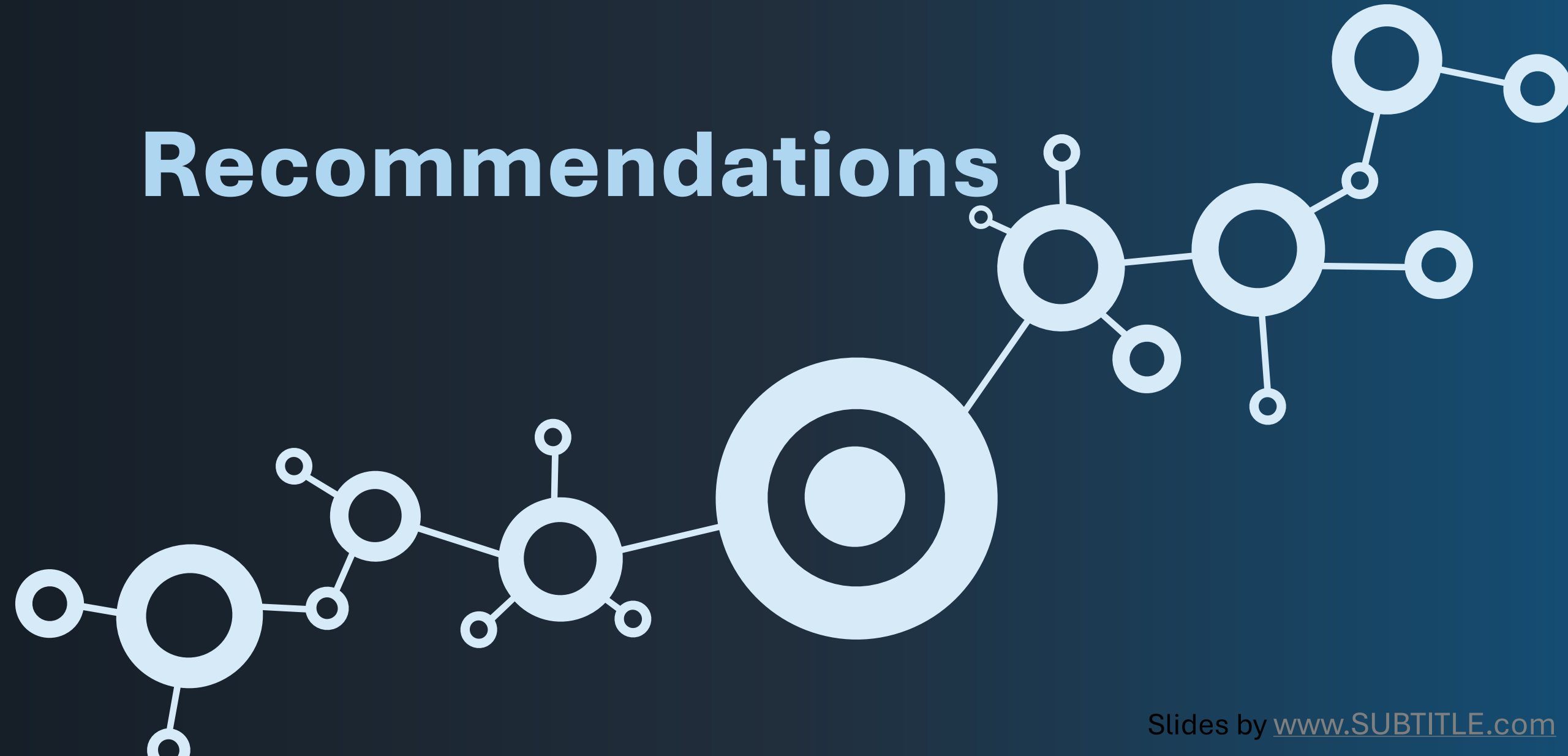
Modern Full Hybrid

Hybrid Functionality

- Outbound HTTPS connections to Exchange Online
 - Exchange Hybrid-Agent Exchange Online to Exchange on-premises communication
 - Exchange Server communication to Exchange Online
 - Install additional Hybrid-Agents for redundancy
 - Communication to Exchange Online, either directly or using outbound proxy servers



Recommendations



Recommendations

- Use Exchange Server SE as an on-premises hybrid endpoint
 - Microsoft Teams backend services use AutoDiscover v2 to discover on-premises Exchange Web Services and REST endpoints
 - Client Access Endpoint must be accessible for Microsoft Teams backend services
 - Always install the latest Exchange Server cumulative update and security updates
- Use Third-Party TLS certificate with all required subject alternate names (SAN) for inbound HTTPS connections
 - AutoDiscover
 - Exchange namespace → Exchange Virtual Directory's external URL-Settings

Recommendations

- Use a dedicated Third-Party TLS certificate for SMTP connections when using centralized mail flow
- Enable and configure OAUTH authentication using Hybrid Configuration Wizard
 - Exchange product group provides detailed information on how to configure OAUTH manually
- Publish AutoDiscover DNS resource records for all SMTP domains used for primary email addresses
- Ensure to activate the "Exchange Hybrid" option in Entra ID Connect (aka Azure AD Connect) before running the Exchange Hybrid Configuration Wizard
 - Without that option activated, you'll risk duplicate mailboxes (On-Premises and Exchange Online)
- Configure and activate the dedicated Entra Hybrid App for On-Premises to Exchange Online authentication
 - Legacy First-Party-App authentication is blocked since October 2025

Q&A



Ressources

- [Exchange Server Hybrid Deployments](#)
- [Hybrid Deployment Prerequisites](#)
- [Hybrid Configuration Wizard FAQs](#)
- [Configuring Teams calendar access for Exchange on-premises mailboxes](#)
- [Configure OAuth authentication between Exchange and Exchange Online organizations](#)
- [Demystifying and troubleshooting hybrid mail flow: when is a message internal?](#)
- [Remote Connectivity Analyzer](#)
- [How Exchange and Microsoft Teams interact](#)
- [Microsoft Teams and on-premises mailboxes](#)



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